

The zero-state tangent cone is the exact obstruction

Conditional reduction for the conjecture in arXiv:1802.04424

11 August 2026

Sharp conditional reduction, not a solution of the conjecture. For a bi-approximately unital L^p -operator algebra, every bounded-coefficient limit of positive multiples of states is automatically nonnegative on the support idempotent. The conjecture is equivalent to the same nonnegativity for the remaining limits, which are precisely weak-star tangent directions at the zero state restriction. Thus a counterexample must exhibit an unboundedly rescaled state net converging to a functional which is negative on the support idempotent.

The question

Blecher and Phillips prove that a real-positive bounded approximate identity implies bi-approximate unitality and conjecture the converse [1].

Lemma 4.4. *Let A be an approximately unital Arens regular Banach algebra. If A has a real positive bounded approximate identity, then A is bi-approximately unital in the sense of Definition 4.2.*

Proof. Lemma 2.22 implies that A has a cai in $\mathfrak{F}_A$. This cai converges weak* to the identity e of A^{**} by Lemma 1.14. Since norm closed balls are weak* closed, we get $\|e\| \leq 1$ and $\|1 - e\| \leq 1$. Hence e is bicontractive. $\square$

We conjecture that the converse of Lemma 4.4 is always true for L^p -operator algebras, namely that a bi-approximately unital L^p -operator algebra A has a real positive cai. Corollary 2.23 may be helpful for this question.

In [Gardella and Thiel 2016] it is shown that quotients of L^p -operator algebras by closed ideals need not be L^p -operator algebras, giving a negative solution to Problem 3.8 in [Le Merdy 1996]. Things are better if the ideal is approximately unital.

Let A be a nonunital approximately unital Arens regular Banach algebra, A^1 its multiplier unitization, and e the identity of A^{**} viewed as the central support idempotent in $(A^1)^{**}$. Write $S(A)$ for the state space used in the source and

$$\mathcal{R}(A) = \{\Phi|_A : \Phi \in S(A^1)\}.$$

By Lemma 2.25(5) of [1], $\mathcal{R}(A)$ is the weak-star closure of $S(A)$ in A^* . The real-positive cone and its dual cone are

$$\mathfrak{r}_A = \{a \in A : \operatorname{Re} \omega(a) \geq 0 \ (\omega \in S(A))\}, \quad \mathfrak{c}_{A^*} = \{f \in A^* : \operatorname{Re} f(a) \geq 0 \ (a \in \mathfrak{r}_A)\}.$$

Define the weak-star tangent cone of the state space at zero by

$$\operatorname{Tan}_{w^*}(S(A), 0) = \left\{ f \in A^* : \begin{array}{l} \text{there are nets } \omega_i \in S(A), \ t_i \geq 0, \ t_i \rightarrow \infty, \\ t_i \omega_i \longrightarrow f \text{ in } \sigma(A^*, A) \end{array} \right\}.$$

Idea of the proof

Real polar duality says that $\mathfrak{c}_{A^*}$ is the weak-star closure of the cone generated by $S(A)$. When $S(A)$ is convex, an element of that cone is simply $t\omega$. Passing to a subnet, the scalar t converges in the compactified half-line $[0, \infty]$. A finite limit produces a ray through an actual limiting state restriction; an infinite limit produces exactly a tangent direction at zero. Bicontractivity handles every finite ray. Bipolarity then shows that the original conjecture is equivalent to controlling only the tangent directions.

Cone decomposition

Theorem 1 (finite rays versus zero-state tangents). *Suppose that $S(A)$ is convex. Then*

$$\mathfrak{c}_{A^*} = \mathbb{R}_+ \mathcal{R}(A) \cup \text{Tan}_{w^*}(S(A), 0). \quad (1)$$

Proof. Lemma 2.25(4) of [1] identifies $\mathfrak{r}_A$ as the real polar of $S(A)$. The real bipolar theorem therefore gives

$$\mathfrak{c}_{A^*} = \overline{\text{cone } S(A)}^{\sigma(A^*, A)}.$$

Convexity of $S(A)$ makes its algebraic conic hull $\mathbb{R}_+ S(A)$. Thus, for $f \in \mathfrak{c}_{A^*}$, there is a net $t_i \omega_i \rightarrow f$ weak-star with $t_i \geq 0$ and $\omega_i \in S(A)$.

The compact space $[0, \infty]$ contains a convergent subnet of (t_i) . If $t_i \rightarrow 0$, then $\|t_i \omega_i\| = t_i \rightarrow 0$, so $f = 0$. If $t_i \rightarrow t \in (0, \infty)$, then $\omega_i \rightarrow f/t$ weak-star, whence $f/t \in \mathcal{R}(A)$. If $t_i \rightarrow \infty$, then the same subnet witnesses $f \in \text{Tan}_{w^*}(S(A), 0)$; indeed $\omega_i = (t_i \omega_i)/t_i \rightarrow 0$ weak-star. This proves one inclusion in (1). Conversely, $\mathbb{R}_+ \mathcal{R}(A)$ lies in the closure of $\mathbb{R}_+ S(A)$ by the definition of $\mathcal{R}(A)$, while every tangent functional lies there by definition. $\square$

Exact obstruction criterion

Theorem 2. *Let $1 < p < \infty$, and let A be a bi-approximately unital L^p -operator algebra. The following are equivalent.*

- (i) *A has a real-positive contractive approximate identity.*
- (ii) *$\text{Re } f^{**}(e) \geq 0$ for every $f \in \text{Tan}_{w^*}(S(A), 0)$.*
- (iii) *There is no net $\omega_i \in S(A)$ and no scalar net $t_i \rightarrow \infty$ for which $t_i \omega_i \rightarrow f$ weak-star and $\text{Re } f^{**}(e) < 0$.*

Proof. The equivalence of (ii) and (iii) is the definition of the tangent cone. Suppose (i) holds, and let (u_j) be the stated cai. Then $u_j \rightarrow e$ weak-star in A^{**} . Every $f \in \mathfrak{c}_{A^*}$ is nonnegative on each $u_j \in \mathfrak{r}_A$, and hence $\text{Re } f^{**}(e) \geq 0$. This gives (ii).

Conversely, assume (ii). The source proves that $S(A)$ is convex (Remark 2.26 and Corollary 4.25(1)). Theorem 1 applies. Since A is bi-approximately unital, $\|1 - e\| \leq 1$; Lemma 2.29(4) of [1] says that the idempotent e is real positive in $(A^1)^{**}$. If $f = t\Phi|_A \in \mathbb{R}_+ \mathcal{R}(A)$, naturality of the bidual restriction gives

$$\text{Re } f^{**}(e) = t \text{Re } \Phi^{**}(e) \geq 0.$$

Hypothesis (ii) gives the same inequality on the tangent part of (1). Hence it holds for every $f \in \mathfrak{c}_{A^*}$. The real bipolar theorem now yields

$$e \in \overline{\mathfrak{r}_A}^{\sigma(A^{**}, A^*)}.$$

For a finite set $F = \{x_1, \dots, x_m\} \subseteq A$, set

$$T_F(a) = (ax_j - x_j, x_j a - x_j)_{j=1}^m \in A^{2m}.$$

Separate weak-star continuity of multiplication gives $0 \in \overline{T_F(\mathfrak{r}_A)}^{\text{weak}}$. The set $T_F(\mathfrak{r}_A)$ is convex, so its weak and norm closures agree. Directing the resulting choices by finite sets and error tolerances produces a real-positive approximate identity, not asserted to be bounded. Since A already has a cai and is Arens regular, Lemma 2.22 of [1] upgrades it to a cai in $\mathfrak{F}_A = \{a \in A : \|1 - a\| \leq 1\} \subseteq \mathfrak{r}_A$. Thus (i) holds. $\square$

Corollary 1. *The conjecture holds for every bi-approximately unital L^p -operator algebra for which each element of $\mathfrak{c}_{A^*}$ admits a weak-star approximating net $t_i\omega_i$ with bounded (t_i) . More generally, it is enough that every zero-state tangent functional be nonnegative on e .*

Why uniform boundedness does not remove the obstruction

Proposition 1. *For every nonunital approximately unital A , there are nets $\omega_i \in S(A)$ and $t_i \rightarrow \infty$ such that $t_i\omega_i \rightarrow 0$ weak-star. If $S(A)$ is convex, the net can instead be chosen so that $t_i\omega_i \rightarrow \omega_0$ for any prescribed $\omega_0 \in S(A)$.*

Proof. The quotient state of A^1 restricts to zero on A , so Lemma 2.25(5) of [1] gives $0 \in \overline{S(A)}^{w*}$. Direct the pairs (F, n) , where $F \subseteq A$ is finite and $n \in \mathbb{N}$, by inclusion and increasing n . Choose $\eta_{F,n} \in S(A)$ such that $|\eta_{F,n}(x)| < n^{-2}$ for every $x \in F$, and set $t_{F,n} = n$. Then $t_{F,n}\eta_{F,n} \rightarrow 0$ weak-star although $\|t_{F,n}\eta_{F,n}\| = n \rightarrow \infty$.

For the final assertion, use convexity and put

$$\omega_{F,n} = n^{-1}\omega_0 + (1 - n^{-1})\eta_{F,n}.$$

Then $\omega_{F,n} \in S(A)$ and $n\omega_{F,n} = \omega_0 + (n-1)\eta_{F,n} \rightarrow \omega_0$ weak-star. $\square$

Thus weak-star convergence of a net in A^* does not imply that the net has a norm-bounded range or even a norm-bounded tail. Banach–Steinhaus applies to pointwise bounded families; convergence of an arbitrary net gives only pointwise eventual boundedness, with the relevant tail depending on the test vector. Proposition 1 shows that the distinction occurs in the exact state space of this problem.

Boundary of the result

Theorem 2 does not establish the sign of a zero-state tangent functional on e , so it does not settle the conjecture. Three natural routes were checked and found to stop at the same boundary:

1. Local reflexivity and the u-ideal condition give approximate elements with $\|1 - 2a\| \leq 1 + \varepsilon$, but an exact real-positive correction is a proximality/Kaplansky-density issue.
2. Hahn–Banach smoothness uniquely identifies norm-preserving extensions, but does not make the u-projection positive on all real-positive functionals.
3. Multiplying an accretive element by the central accretive support idempotent is not a safe positivity argument: even commuting real-positive idempotents in an L^p -operator algebra can have non-real-positive product (Example 3.3 of [1]).

Consequently, a proof must control the tangent cone by additional L^p -geometry, while a counterexample must realize a negative tangent functional as in Theorem 2(iii). This is the precise point left open by the present reduction.

Bounded novelty check

On 11 August 2026, the exact conjecture, its end-of-paper restatement, and later papers citing the source were checked in the run indexes and by bounded exact-phrase web/arXiv searches. No later resolution or this tangent-cone formulation was located. This is evidence for novelty, not a guarantee.

References

- [1] D. P. Blecher and N. C. Phillips, *L^p -operator algebras with approximate identities, I*, Pacific J. Math. **303** (2019), 401–457; arXiv:1802.04424.
- [2] D. P. Blecher and N. Ozawa, *Real positivity and approximate identities in Banach algebras*, Pacific J. Math. **277** (2015), 1–59.